



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

June 2011
Assessment Task 3
Year 12

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question.
- All answers to be given in simplified exact form unless otherwise stated.

Total Marks – 92

- Attempt questions 1-6
- All questions are of equal value.
- Start each new question in a separate answer booklet.
- Hand in your answers in 4 separate bundles: Section A (Questions 1 and 2), Section B (Questions 3 and 4, Section C (Questions 5 and 6)

Examiner: *A Ward*

Start a new booklet.

Section A

Question 1 (15 marks). **Marks**

a) In how many ways can 8 people sit at a round table? 1

b) For the polynomial $(a-6)x^5 + (2b+1)x^2 + (12-4c)$, state the possible value(s) of a , b and c if the polynomial is: 5

(i) monic of degree 5.

(ii) of degree 2.

(iii) a zero polynomial.

c) Find $\frac{dy}{dx}$ given the following: 9

(i) $y = \frac{1}{x+3}$

(ii) $y = \sin(1+x^3)$

(iii) $y = e^{4x+2}$

(iv) $y = \frac{\ln(x+2)}{\ln(x^2+1)}$

End of Question 1

Question 2 (15 Marks).**Marks**

- a)**
- (i) Given $f(x) = \frac{5-3x}{4}$, find $f^{-1}(x)$ 3
- (ii) Sketch the graphs $f(x)$ and $f^{-1}(x)$ on the same axes.
- b)** Find the following: 4
- (i) $\int \frac{6}{3x+1} dx$
- (ii) $\int \frac{e^x + 1}{e^x} dx$
- c)** Evaluate: $\int_1^2 \frac{3x^2 - 5x + 7}{x} dx$ 3
- d)** Find an exact value for $\sin 15^\circ$, showing all working. 3
- e)** If $y = \sin^{-1}(x)$, find $\frac{d^2y}{dx^2}$. 2

End of Question 2**End of Section A**

Section B – Start a new booklet.**Question 3 (15 Marks).****Marks**

- a)** $Q(x) = ax^2 + bx + c$ and when $Q(x)$ is divided by $(x - m)$ or $(x - n)$, the remainders are the same. Prove that, if $m \neq n$ then $(m + n)$ is equal to the sum of the roots of $Q(x)$. 4
- b)** (i) What are the range and period of $y = 2 \cos\left(\frac{x}{2}\right)$? 5
- (ii) Draw the graph of $y = 1 - 2 \cos\left(\frac{x}{2}\right)$ for $-2\pi \leq x \leq 2\pi$.
- c)** (i) In how many ways can 10 individuals be divided into 2 groups of 5? 6
- (ii) If the two youngest individuals must be in the same group, how many ways are there now?
- (iii) How many ways are there, if the two youngest individuals must not be in the same group?

End of Question 3

Question 4 (15 Marks).**Marks**

- a) (i) Express $\sqrt{3} \cos 2t - \sin 2t$ in the form $A \cos(2t + \alpha)$ with $A > 0$ and

4

$$0 \leq \alpha \leq \frac{\pi}{2}.$$

- (ii) Find, in exact form, the general solution to $\sqrt{3} \cos 2t - \sin 2t = 1$

- b) Show that:

3

$$\frac{n!}{(n-r)!r!} + \frac{n!}{(n-r+1)!(r-1)!} = \frac{(n+1)!}{r!(n-r+1)!}$$

- c) For $n = 1, 2$ and 3 , find the value of :

4

$$\int_{\frac{\pi}{2}}^{\pi} \cos nx \, dx$$

- d) (i) Find the second derivative of $y = \ln(x^2 + 4)$

4

- (ii) .Write the equation of the tangent to $y = x \tan^{-1}\left(\frac{x}{2}\right)$ at $x=2$

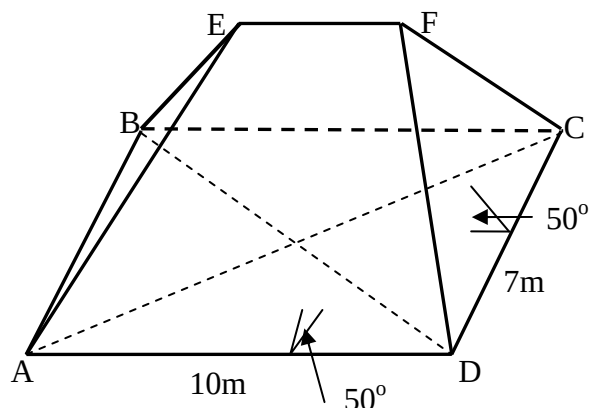
End of Question 4**End of Section B**

Section C – Start a new booklet.

Question 5 (15 Marks).

Marks

a)



A roof of a house has rectangular base 7m by 10m. Each of the four sloping faces makes an angle of 50° with the horizontal.

5

- (i) Calculate the length of the ridge EF
- (ii) Calculate the total surface area of the sloping faces to 1 decimal place.

b) Prove :

4

$$\sec 2A + \tan 2A \equiv \frac{\cos A + \sin A}{\cos A - \sin A}$$

c) Use the substitution $u = 2 \sin x$ to find:

3

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{8 \cos x}{1 + 4 \sin^2 x} dx$$

d) Factorise $24x^3 - 14x^2 - 63x + 45$, given that one of the zeros is $\frac{3}{2}$.

3

End of Question 5

Question 6 (17 marks).**Marks**

- a) Solve the following for $0 \leq t \leq 2\pi$:

2

$$\sin 2t + 4 \sin t + 2 \cos t = -4$$

- b) If $t = \tan\left(\frac{\theta}{2}\right)$, express in terms of t : $\frac{1 - 2 \sin \theta}{2 \cos \theta + 1}$

3

Leave your answer in simplified exact form.

- c) (i) Find the coordinates of P , the point of intersection of the curves:

5

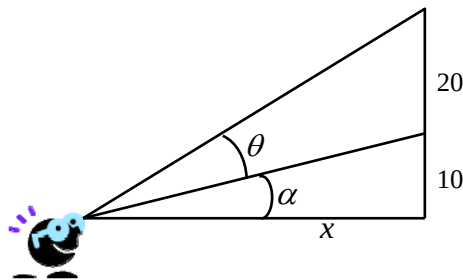
$$y = e^x$$

$$y = 2 + 3e^{-x}$$

- (ii) If these curves cut the y -axis at the points A and B . Calculate the area bounded by AB and the arcs AP and BP .

- d) A movie screen on a wall is 20 metres high and 10 metres above eye level. You are x metres from the screen.

7



- (i) Show that $\theta = \tan^{-1}\left(\frac{30}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right)$

- (ii) Find $\lim_{x \rightarrow 0} \left(\tan^{-1}\left(\frac{30}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right) \right)$

- (iii) Find $\lim_{x \rightarrow \infty} \left(\tan^{-1}\left(\frac{30}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right) \right)$

- (iv) At what distance should you position yourself so that the viewing angle θ of the movie screen is as large as possible?

End of Question 6.

End of Section C.

End of Examination.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

1 (a)

7!

$$(6) \quad (i) \quad (a-b) = 1$$

$$a = 7$$

$$(ii) \quad a - b = 0$$

$$a = 6$$

$$(iii) \quad a = 6, b = -\frac{1}{2}, c = 3, 3$$

$$(c) \quad (i) \quad y = (x+3)^{-1}$$

$$\frac{dy}{dx} = -(x+3)^{-2} \cdot 1$$

$$= -\frac{1}{(x+3)^2}$$

$$(ii) \quad y = \sin(1+x^3)$$

$$\frac{dy}{dx} = \cos(1+x^3) \cdot 3x^2$$

$$= 3x^2 \cos(1+x^3)$$

$$(iii) \quad y = e^{4x+2}$$

$$\frac{dy}{dx} = e^{4x+2} \cdot 4$$

$$= 4e^{4x+2}$$

$$(iv) \quad y = \frac{\ln(x+2)}{\ln(x^2+1)}$$

$$\frac{dy}{dx} = \frac{v \frac{dv}{dx} - u \frac{du}{dx}}{v^2}$$

$$= \frac{\ln(x^2+1) \cdot \frac{1}{x^2+1} - \ln(x+2) \cdot \frac{2x}{x^2+1}}{[\ln(x^2+1)]^2}$$

$$= \frac{\ln(x^2+1) - 2x \ln(x+2)}{(x^2+1) [\ln(x^2+1)]^2}$$

$$= \frac{(x^2+1) \ln(x^2+1) - 2x(x+2) \ln(x+2)}{(x+2)(x^2+1) [\ln(x^2+1)]^2}$$

$$2 (a) \quad f(x) = \frac{5-3x}{4}; \quad x \in \mathbb{R}, y \in \mathbb{R}$$

$$y = \frac{5-3x}{4}$$

$$y = \frac{5}{4} - \frac{3}{4}x$$

For $f^{-1}(x)$

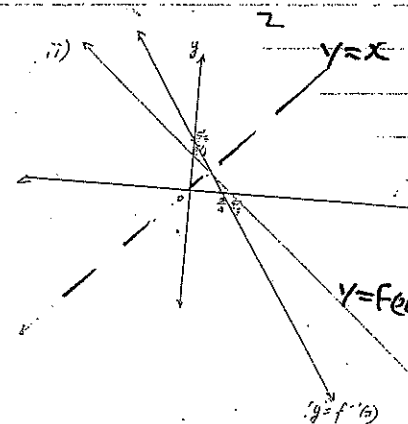
$$x = \frac{5}{4} - \frac{3}{4}y$$

$$4x = 5 - 3y$$

$$3y = 5 - 4x$$

$$y = \frac{5}{3} - \frac{4}{3}x$$

$$f^{-1}(x) = \frac{5}{3} - \frac{4}{3}x$$



$$(b) \quad (i) \quad \int \frac{6}{3x+1} \cdot dx$$

$$= 2 \int \frac{3}{3x+1} \cdot dx$$

$$= 2 \ln(3x+1) + C$$

$$(ii) \quad \int \frac{e^x+1}{e^x} \cdot dx$$

$$= \int 1 + e^{-x} \cdot dx$$

$$= x - e^{-x} + C$$

$$(c) \quad \int \frac{3x^2 - 5x + 7}{x} \cdot dx$$

$$= \int 3x - 5 + \frac{7}{x} \cdot dx$$

$$= \left[\frac{3x^2}{2} - 5x + 7 \ln x \right]_1^2$$

$$= \left(\frac{3}{2} - 10 + 7 \ln 2 \right) - \left(\frac{3}{2} - 5 + 7 \ln 1 \right)$$

$$= 7 \ln 2 - 4 + 3 \frac{1}{2}$$

$$= 7 \ln 2 - \frac{1}{2}$$

15

(d)

$$y = \sin 15^\circ$$

$$y = \sin (45 - 30)^\circ \quad (1)$$

$$= \sin 45 \cos 30 - \cos 45 \sin 30$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

(2)

(e)

$$y = \sin^{-1}(x)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$= (1-x^2)^{-\frac{1}{2}}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{2} (1-x^2)^{-\frac{3}{2}} \cdot (-2x) \quad (1)$$

$$= \frac{x}{\sqrt{(1-x^2)^3}}$$

(1)

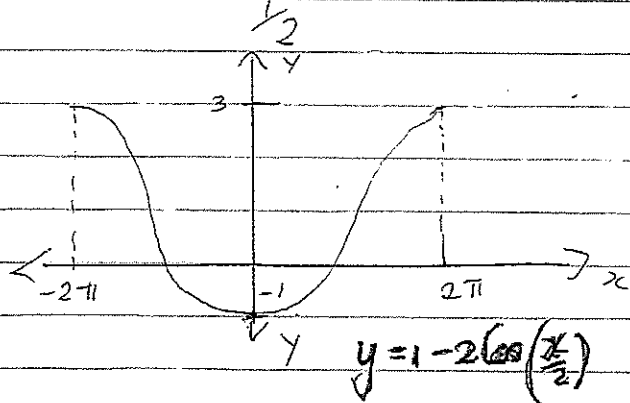
15

QUESTION 3

(a) $Q(m) = Q(n)$
 $am^2 + bm + c = an^2 + bn + c$
 $a(m^2 - n^2) = -b(m - n)$
 $a(m+n) = -b$
 $m+n = \frac{-b}{a}$
 = Sum of roots.

(b) Range $-2 \leq y \leq 2$

Period $= \frac{2\pi}{\frac{1}{2}} = 4\pi$



(c) (i) 10
 $C_5 \div 2 = 126$

(ii) ${}^8C_3 = 56$

(iii) ${}^8C_4 = 70$
 or (i) - (ii)

4(a) (ii)

$2 \cos(2t + \pi/6) = 1$
 $\cos(2t + \pi/6) = \frac{1}{2}$
 $2t + \pi/6 = \pm \pi/3 + 2n\pi$
 $2t = -\pi/6 \pm \pi/3 + 2n\pi$
 $t = n\pi - \pi/4 \pm \pi/6$

QUESTION 4

(a) $\sqrt{3} \cos t - \sin 2t = A \cos(2t + \alpha)$
 $= A(\cos 2t \cos \alpha - \sin 2t \sin \alpha)$
 $A \cos \alpha = \sqrt{3}$
 $A \sin \alpha = 1$
 $A = \sqrt{3+1} = 2$
 $\cos A = \sqrt{3}/2 \quad \sin A = \frac{1}{2} \quad \alpha = \pi/6$
 $\sqrt{3} \cos t - \sin 2t = 2 \cos(2t + \pi/6)$

(b) $\frac{n!}{(n-r)!r!} + \frac{n!}{(n-r+1)!(r-1)!} = \frac{n!(n-r+1) + n!(r)}{(n-r+1)!r!}$

$= \frac{n!(n-r+1-r)}{(n-r+1)!r!} = \frac{n+1!}{(n-r+1)!r!}$

(c) $n=1 \quad \sin \pi - \sin \pi/2 = 0 - 1 = -1$
 $n=2 \quad \frac{1}{2} \sin 2\pi - \frac{1}{2} \sin \pi = 0 - 0 = 0$
 $n=3 \quad \frac{1}{3} \sin 3\pi - \frac{1}{3} \sin \frac{3\pi}{2} = 0 - (-\frac{1}{3}) = \frac{1}{3}$

(d) (i) $y = \ln(x^2 + 4)$
 $y' = \frac{2x}{x^2 + 4}$
 $y'' = \frac{(x^2 + 4)^2 - 2x \times 2x}{(x^2 + 4)^2}$
 $= \frac{-2x^2 + 8}{(x^2 + 4)^2}$

(ii) $y = x \tan^{-1} \frac{x}{2}$
 $y' = \tan^{-1} \frac{x}{2} \times 1 + x \times \frac{2}{4+x^2}$
 $= \tan^{-1} \frac{x}{2} + \frac{2x}{4+x^2}$

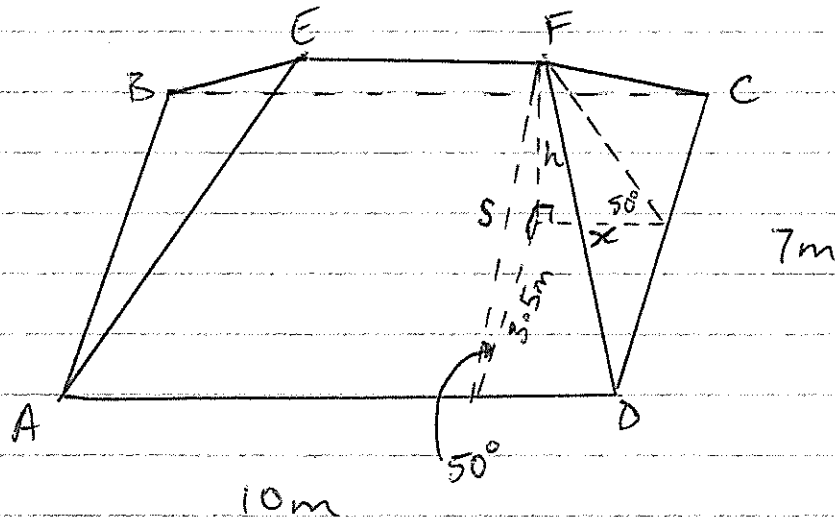
$x=2 \quad m = \frac{\pi}{4} + \frac{1}{2} \quad y = \frac{\pi}{2}$

eqn $y - \frac{\pi}{2} = \left(\frac{\pi}{4} + \frac{1}{2}\right)(x - 2)$

$y = x\left(\frac{\pi}{4} + \frac{1}{2}\right) - \frac{\pi}{2} - 1 + \frac{\pi}{2}$
 $y = x\left(\frac{\pi}{4} + \frac{1}{2}\right) - 1$

Question 5

a)



$$i) \tan 50^\circ = \frac{h}{3.5}$$

$$h = 3.5 \tan 50^\circ$$

$$\tan 50^\circ = \frac{h}{x}$$

$$x = \frac{3.5 \tan 50^\circ}{\tan 50^\circ}$$

$$x = 3.5$$

$$EF = 10 - 2(3.5)$$

$$= 3 \text{ m}$$

$$ii) \cos 50^\circ = \frac{3.5}{s}$$

$$s = \frac{3.5}{\cos 50^\circ}$$

$$S.A = 2 \times \left[\frac{1}{2} \times 7 \times \frac{3.5}{\cos 50^\circ} \right] + 2 \times \left[\frac{3+10}{2} \times \frac{3.5}{\cos 50^\circ} \right]$$

$$= 108.9 \text{ m}^2 \text{ (to 1 decimal place)}$$

$$\begin{aligned}
 \text{b) } LHS &= \sec 2A + \tan 2A \\
 &= \frac{1}{\cos 2A} + \frac{\sin 2A}{\cos 2A} \\
 &= \frac{1 + \sin 2A}{\cos 2A} \\
 &= \frac{1 + 2\sin A \cos A}{\cos^2 A - \sin^2 A} \\
 &= \frac{\sin^2 A + \cos^2 A + 2\sin A \cos A}{\cos^2 A - \sin^2 A} \\
 &= \frac{(\cos A + \sin A)^2}{(\cos A - \sin A)(\cos A + \sin A)} \\
 &= \frac{\cos A + \sin A}{\cos A - \sin A} \\
 &= RHS
 \end{aligned}$$

$$\text{c) } \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{8 \cos x}{1 + 4 \sin^2 x} dx$$

$$u = 2 \sin x$$

$$\frac{du}{dx} = 2 \cos x$$

$$dx = \frac{du}{2 \cos x}$$

$$= \int_1^{\sqrt{3}} \frac{8 \cos x}{1 + u^2} \cdot \frac{du}{2 \cos x}$$

limit change

$$x = \frac{\pi}{6}$$

$$x = \frac{\pi}{3}$$

$$u = 2 \sin \frac{\pi}{6}$$

$$u = 2 \sin \frac{\pi}{3}$$

$$u = 1$$

$$u = \sqrt{3}$$

$$= 4 \left[\tan^{-1} u \right]_1^{\sqrt{3}}$$

$$= 4 \left[\tan^{-1} \sqrt{3} - \tan^{-1} 1 \right]$$

$$= 4 \left[\frac{\pi}{3} - \frac{\pi}{4} \right]$$

$$= \frac{\pi}{3}$$

d) $P(x) = 24x^3 - 14x^2 - 63x + 45$

If $\frac{3}{2}$ is a zero, $(2x-3)$ is a factor of $P(x)$.

$$\begin{array}{r}
 12x^2 + 11x - 15 \\
 2x-3 \overline{) 24x^3 - 14x^2 - 63x + 45} \\
 \underline{24x^3 - 36x^2} \quad \downarrow \\
 22x^2 - 63x \quad \downarrow \\
 \underline{22x^2 - 33x} \quad \downarrow \\
 -30x + 45 \\
 \underline{-30x + 45} \\
 0
 \end{array}$$

$$P(x) = (2x-3)(12x^2 + 11x - 15)$$

$$= (2x-3)(12x+20)(12x-9)$$

$$\begin{array}{c} 12 \\ 4 \times 3 \end{array}$$

$$= (2x-3)(3x+5)(4x-3)$$

$$\begin{array}{r}
 \times \quad | -180 \\
 + \quad | 11 \\
 \hline
 20, -9
 \end{array}$$

Question 6

a) $\sin 2t + 4\sin t + 2\cos t = -4$

$$2\sin t \cos t + 4\sin t + 2\cos t + 4 = 0$$

$$2\sin t (\cos t + 2) + 2(\cos t + 2) = 0$$

$$(\cos t + 2)(2\sin t + 2) = 0$$

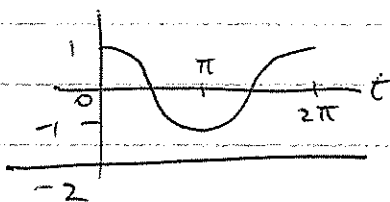
$$\cos t + 2 = 0$$

$$\cos t = -2$$

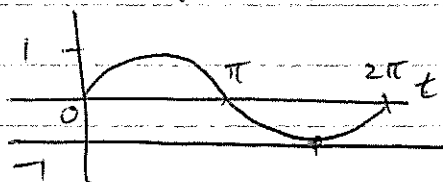
$$2\sin t + 2 = 0$$

$$2\sin t = -2$$

$$\sin t = -1$$

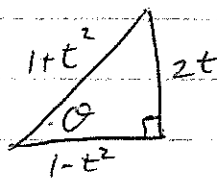
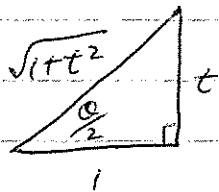


No solution



$$t = \frac{3\pi}{2}$$

b) $t = \tan\left(\frac{\theta}{2}\right)$



$$\frac{1 - 2\sin\theta}{2\cos\theta + 1} = \frac{1 - 2\left(\frac{2t}{1+t^2}\right)}{2\left(\frac{1-t^2}{1+t^2}\right) + 1} \times \frac{1+t^2}{1+t^2}$$

$$= \frac{1+t^2-4t}{2-2t^2+1+t^2}$$

$$= \frac{t^2-4t+1}{3-t^2}$$

c) i) $y = e^x$ — (1)

$y = 2 + 3e^{-x}$ — (2)

sub. (1) into (2)

$$e^x = 2 + 3e^{-x}$$

multiply both sides by e^x

$$(e^x)^2 = 2e^x + 3$$

$$(e^x)^2 - 2e^x - 3 = 0$$

$$\begin{array}{r} x \mid -3 \\ + \underline{-2} \\ -3 \end{array}$$

$$(e^x - 3)(e^x + 1) = 0$$

$$e^x = 3 \quad \text{or} \quad e^x = -1$$

$$x = \ln 3$$

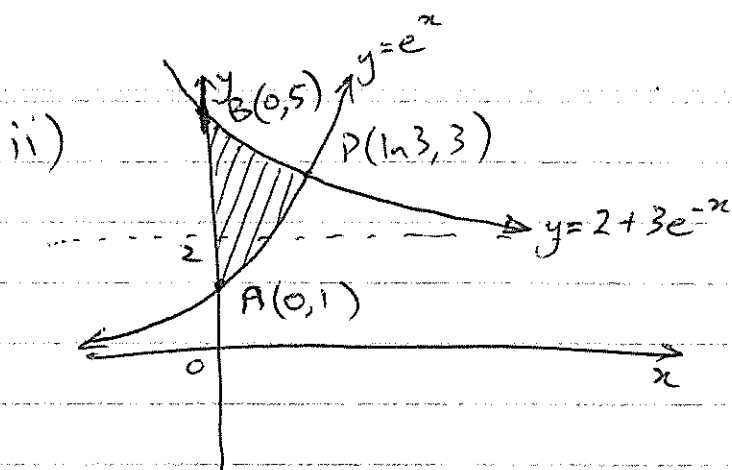
No solution since $e^x > 0$.

sub. into (1)

$$y = e^{\ln 3}$$

$$y = 3$$

$\therefore P$ has coordinates $(\ln 3, 3)$



$$A = \int_0^{\ln 3} (2 + 3e^{-x} - e^x) dx$$

$$= \left[2x - 3e^{-x} - e^x \right]_0^{\ln 3}$$

$$= 2\ln 3 - 3e^{-\ln 3} - e^{\ln 3} - (2(0) - 3e^{-(0)} - e^0)$$

$$= 2\ln 3 - 3e^{\ln(\frac{1}{3})} - 3 - (-3 - 1)$$

$$= 2\ln 3 - 3(\frac{1}{3}) + 1$$

$$= 2\ln 3 \text{ units}^2.$$

d) i) $\tan(\theta + \alpha) = \frac{30}{\pi}$

$$\theta + \alpha = \tan^{-1}\left(\frac{30}{\pi}\right)$$

$$\tan(\alpha) = \frac{10}{\pi}$$

$$\alpha = \tan^{-1}\left(\frac{10}{\pi}\right)$$

$$\theta = (\theta + \alpha) - \alpha$$

$$= \tan^{-1}\left(\frac{30}{\pi}\right) - \tan^{-1}\left(\frac{10}{\pi}\right)$$

ii) as $x \rightarrow 0$

$$\frac{30}{x} \rightarrow \infty$$

$$\tan^{-1}\left(\frac{30}{x}\right) \rightarrow \frac{\pi}{2}$$

$$\text{Similarly, } \tan^{-1}\left(\frac{10}{x}\right) \rightarrow \frac{\pi}{2}$$

$$\therefore \lim_{x \rightarrow 0} \left(\tan^{-1}\left(\frac{30}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right) \right)$$

$$= \frac{\pi}{2} - \frac{\pi}{2}$$

$$= 0$$

iii) as $x \rightarrow \infty$

$$\frac{30}{x} \rightarrow 0$$

$$\tan^{-1}\left(\frac{30}{x}\right) \rightarrow 0$$

$$\text{Similarly, } \tan^{-1}\left(\frac{10}{x}\right) \rightarrow 0$$

$$\therefore \lim_{x \rightarrow \infty} \left(\tan^{-1}\left(\frac{30}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right) \right)$$

$$= 0 - 0$$

$$= 0$$

$$\text{iv) } \theta = \tan^{-1}\left(\frac{30}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right)$$

$$\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{30}{x}\right)^2} \cdot \frac{-30}{x^2} - \frac{1}{1 + \left(\frac{10}{x}\right)^2} \cdot \frac{-10}{x^2}$$

$$= \frac{-30}{x^2 + 900} + \frac{10}{x^2 + 100}$$

$$= \frac{-30(x^2 + 100) + 10(x^2 + 900)}{(x^2 + 900)(x^2 + 100)}$$

For stationary points let $\frac{dQ}{dx} = 0$.

$$-30x^2 - 3000 + 10x^2 + 9000 = 0$$

$$-20x^2 + 6000 = 0$$

$$20x^2 = 6000$$

$$x^2 = 300$$

$$x = \pm\sqrt{300}$$

$$x = 10\sqrt{3} \text{ m} \quad \text{since } x \text{ is a distance}$$

Q is continuous for $x > 0$.

From the limits in (ii) & (iii)

Q is a maximum when $x = 10\sqrt{3} \text{ m}$.